

One of Ramanujan Identity

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1 Ramanujan Identity

Prove that:

$$\alpha \sum_{k=1}^{\infty} \frac{k}{e^{2\alpha k} - 1} + \beta \sum_{k=1}^{\infty} \frac{k}{e^{2\beta k} - 1} = \frac{\alpha + \beta}{24} - \frac{1}{4}$$

$$\alpha > 0, \beta > 0, \alpha \cdot \beta = \pi^2$$

$$f(t) = \begin{cases} \frac{|t|}{e^{2\alpha|t|} - 1} & t \neq 0 \\ \frac{1}{2\alpha} & t = 0 \end{cases}$$

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(t) e^{-i2\pi st} dt = \int_{-\infty}^{\infty} f(t) \cos(2\pi st) dt + i \int_{-\infty}^{\infty} f(t) \sin(2\pi st) dt = 2 \int_0^{\infty} \frac{t \cos(2\pi st)}{e^{2\alpha t} - 1} dt \text{ and } s \neq 0$$

$$\hat{f}(0) = 2 \int_0^{\infty} \frac{t}{e^{2\alpha t} - 1} dt = \frac{2}{(2\alpha)^2} \zeta(2) = \frac{\pi^2}{12\alpha^2} \text{ (Insert Reference)}$$

Notice both the function and it's Fourier Transform are even.
By Poisson Summation formula (Include Reference):

$$\hat{f}(0) + 2 \sum_{m=1}^{\infty} \hat{f}(m) = f(0) + 2 \sum_{n=1}^{\infty} f(n)$$

Hence applying this:

$$\frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + 4 \sum_{m=1}^{\infty} \int_0^{\infty} \frac{t \cos(2\pi mt)}{e^{2\alpha t} - 1} dt$$

Examining

$$\int_0^{\infty} \frac{t \cos(2\pi mt)}{e^{2\alpha t} - 1} dt$$

By considering

$$\int_0^{\infty} \frac{\sin(cx)}{e^x - 1} dx = \int_0^{\infty} \frac{e^{-x} \sin(cx)}{1 - e^{-x}} dx = \int_0^{\infty} \sin cx \sum_{k=1}^{\infty} e^{-kx} dx$$

By the Dominant Convergence Theorem: (Include Reference)

$$\int_0^\infty \sin(cx) \sum_{k=1}^\infty e^{-kx} dx = \sum_{k=1}^\infty \int_0^\infty \sin(cx) e^{-kx} dx$$

By the LaPlace Transform of $\sin at$:

$$\int_0^\infty \sin(ct) e^{-sx} dt = \frac{c}{c^2 + s^2} \text{ (Insert Reference)}$$

$$\sum_{k=1}^\infty \int_0^\infty \sin(cx) e^{-kx} dx = \sum_{k=1}^\infty \frac{c}{c^2 + k^2}$$

By the Tangent Identity (Insert Reference)

$$\sum_{n=-\infty}^\infty \frac{1}{\alpha^2 + n^2} = \frac{\pi}{\alpha} \coth(\pi\alpha)$$

Where

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} = \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{e^{2x} + 1}{e^{2x} - 1}$$

Hence:

$$\sum_{n=-\infty}^\infty \frac{1}{\alpha^2 + n^2} = \frac{\pi}{\alpha} \cdot \frac{e^{2\pi\alpha} + 1}{e^{2\pi\alpha} - 1}$$

Hence:

$$\begin{aligned} \sum_{n=1}^\infty \frac{1}{\alpha^2 + n^2} &= \frac{1}{2} \cdot \left(\frac{\pi}{\alpha} \cdot \frac{e^{2\pi\alpha} + 1}{e^{2\pi\alpha} - 1} - \frac{1}{\alpha^2} \right) \\ &=> \sum_{k=1}^\infty \frac{c}{c^2 + k^2} = \frac{\pi}{2} \cdot \frac{e^{2\pi c} + 1}{e^{2\pi c} - 1} - \frac{1}{2c} \\ &=> \int_0^\infty \frac{\sin(cx)}{e^x - 1} dx = \frac{\pi}{2} \cdot \frac{e^{2\pi c} + 1}{e^{2\pi c} - 1} - \frac{1}{2c} \end{aligned}$$

Taking the derivative of both sides using Leibniz Rule

$$\begin{aligned} \frac{d}{dc} \int_0^\infty \frac{\sin(cx)}{e^x - 1} dx &= \int_0^\infty \frac{\partial}{\partial c} \frac{\sin(cx)}{e^x - 1} dx = \int_0^\infty \frac{x \cos(cx)}{e^x - 1} dx \\ &= \frac{d}{dc} \left(\frac{\pi}{2} \cdot \frac{e^{2\pi c} + 1}{e^{2\pi c} - 1} - \frac{1}{2c} \right) = \frac{\pi}{2} \cdot \frac{2\pi e^{2\pi c}}{e^{2\pi c} - 1} - \frac{\pi}{2} \cdot \frac{2\pi e^{4\pi c} + 2\pi e^{2\pi c}}{(e^{2\pi c} - 1)^2} + \frac{1}{2c^2} = \frac{-2\pi^2 e^{2\pi c}}{(e^{2\pi c} - 1)^2} + \frac{1}{2c^2} \end{aligned}$$

Therefore making a change of variables $x = 2\alpha t$

$$\int_0^\infty \frac{x \cos(cx)}{e^x - 1} dx = \int_0^\infty \frac{2\alpha t \cos(2\alpha ct)}{e^{2\alpha t} - 1} 2\alpha dt = (2\alpha)^2 \int_0^\infty \frac{t \cos(2\alpha ct)}{e^{2\alpha t} - 1} dt = \frac{-2\pi^2 e^{2\pi c}}{(e^{2\pi c} - 1)^2} + \frac{1}{2c^2}$$

Letting $c = \pi \cdot \frac{m}{\alpha}$

$$(2\alpha)^2 \int_0^\infty \frac{t \cos(2\pi m \cdot t)}{e^{2\alpha t} - 1} dt = \frac{-2\pi^2 e^{2\pi^2 \cdot \frac{m}{\alpha}}}{(e^{2\pi^2 \cdot \frac{m}{\alpha}} - 1)^2} + \frac{1}{2\pi^2 \cdot \frac{m^2}{\alpha^2}} = \frac{-2\pi^2 e^{2\pi^2 \cdot \frac{m}{\alpha}}}{(e^{2\pi^2 \cdot \frac{m}{\alpha}} - 1)^2} + \frac{\alpha^2}{2\pi^2 m^2}$$

Plugging this result back into our original statement:

$$\frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty} \left(\frac{-2\pi^2 e^{2\pi^2 \cdot \frac{m}{\alpha}}}{\alpha^2 (e^{2\pi^2 \cdot \frac{m}{\alpha}} - 1)^2} + \frac{1}{2\pi^2 m^2} \right) = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty} \frac{-2\pi^2 e^{2\pi^2 \cdot \frac{m}{\alpha}}}{\alpha^2 (e^{2\pi^2 \cdot \frac{m}{\alpha}} - 1)^2} + \frac{1}{2\pi^2} \zeta(2)$$

$$\frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty} \frac{-2\pi^2 e^{2\pi^2 \cdot \frac{m}{\alpha}}}{\alpha^2 (e^{2\pi^2 \cdot \frac{m}{\alpha}} - 1)^2} + \frac{1}{12}$$

$$\text{Let } \beta = \frac{\pi^2}{\alpha}$$

$$\Rightarrow \frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + \frac{1}{12} - \frac{2\pi^2}{\alpha^2} \sum_{m=1}^{\infty} \frac{e^{2m\beta}}{(e^{2m\beta} - 1)^2}$$

$$\text{Examining } \sum_{m=1}^{\infty} \frac{e^{2m\beta}}{(e^{2m\beta} - 1)^2}$$

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} &= \sum_{n=1}^{\infty} \frac{n e^{-2\beta n}}{1 - e^{-2\beta n}} = \sum_{n=1}^{\infty} \left(n \sum_{m=1}^{\infty} e^{-2\beta \cdot n \cdot m} \right) \\ \sum_{n=1}^{\infty} \left(n \sum_{m=1}^{\infty} e^{-2\beta \cdot n \cdot m} \right) &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} n e^{-2\beta \cdot n \cdot m} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} n e^{-2\beta \cdot n \cdot m} \end{aligned}$$

Insert Reference from Geometric Series Section

$$\sum_{m=1}^{\infty} \frac{e^{-2\beta \cdot m}}{(1 - e^{-2\beta \cdot m})^2} = \sum_{m=1}^{\infty} \frac{e^{2\beta \cdot m}}{(e^{2\beta \cdot m} - 1)^2} = \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1}$$

Hence combining this fact we obtain:

$$\begin{aligned} \Rightarrow \frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} &= \frac{\pi^2}{12\alpha^2} + \frac{1}{12} - \frac{2\pi^2}{\alpha^2} \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} \\ \frac{1}{2\alpha} + 2 \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} &= \frac{\beta}{12\alpha} + \frac{1}{12} - \frac{2\beta}{\alpha} \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} \\ \frac{1}{2} + 2\alpha \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} &= \frac{\beta}{12} + \frac{\alpha}{12} - 2\beta \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} \end{aligned}$$

$$\boxed{\alpha \sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} + \beta \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} = \frac{\beta + \alpha}{24} - \frac{1}{4}}$$

Where $\alpha \cdot \beta = \pi^2$

The special case of $\alpha = \beta = \pi$:

$$\boxed{\sum_{n=1}^{\infty} \frac{n}{e^{2\pi n} - 1} = \frac{1}{24} - \frac{1}{8\pi}}$$